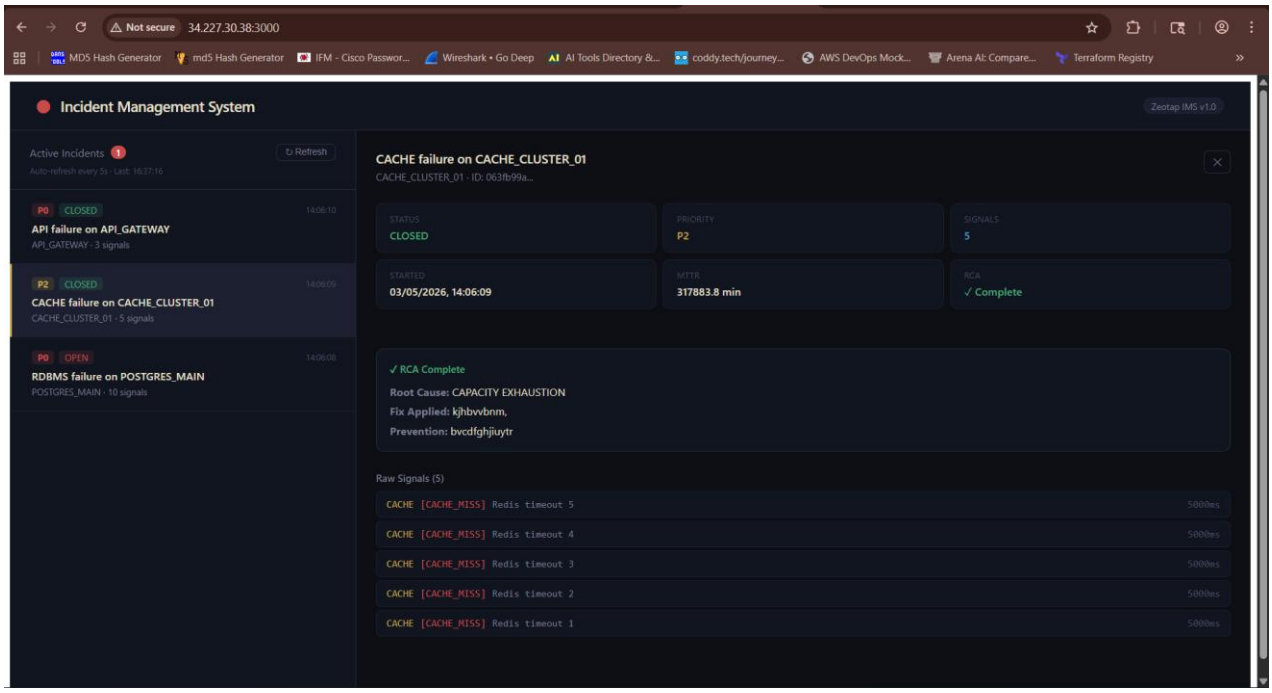
**MHA Ramayya – Infrastructure / SRE Intern Assignment**

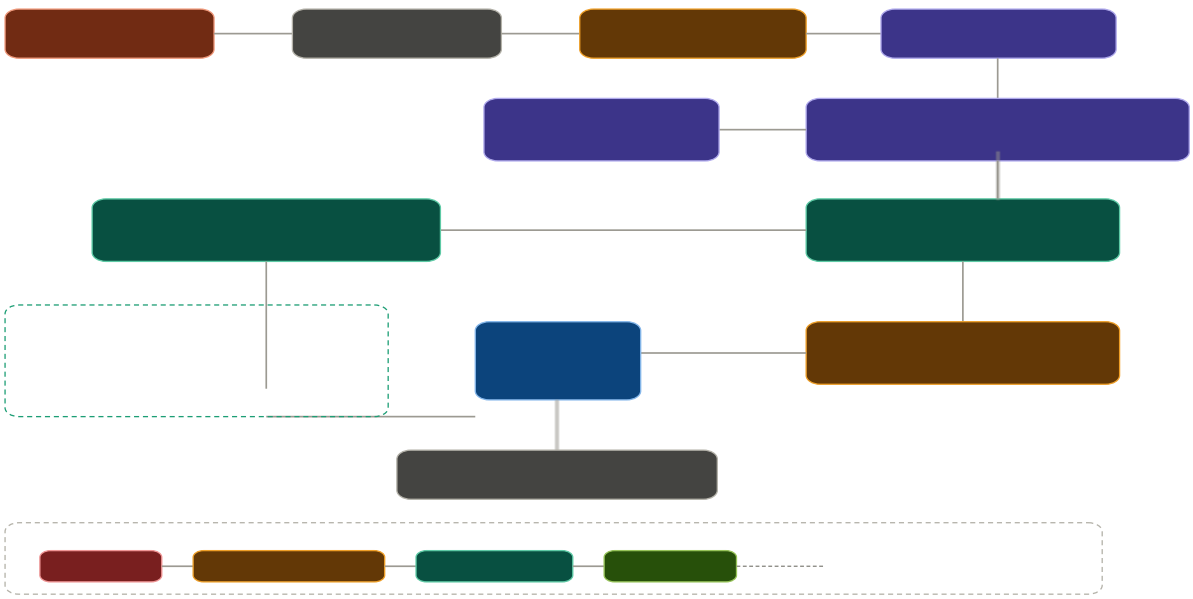



## 1. Project Overview

This project implements a **production-style Incident Management System (IMS)** with a complete DevOps lifecycle: CI/CD, monitoring, observability, and containerization. It demonstrates automation, resilience, and scalable deployment practices.

**2. GitHub Repository** https://github.com/babai-cyber/ims.git

### 3. Architecture Diagram



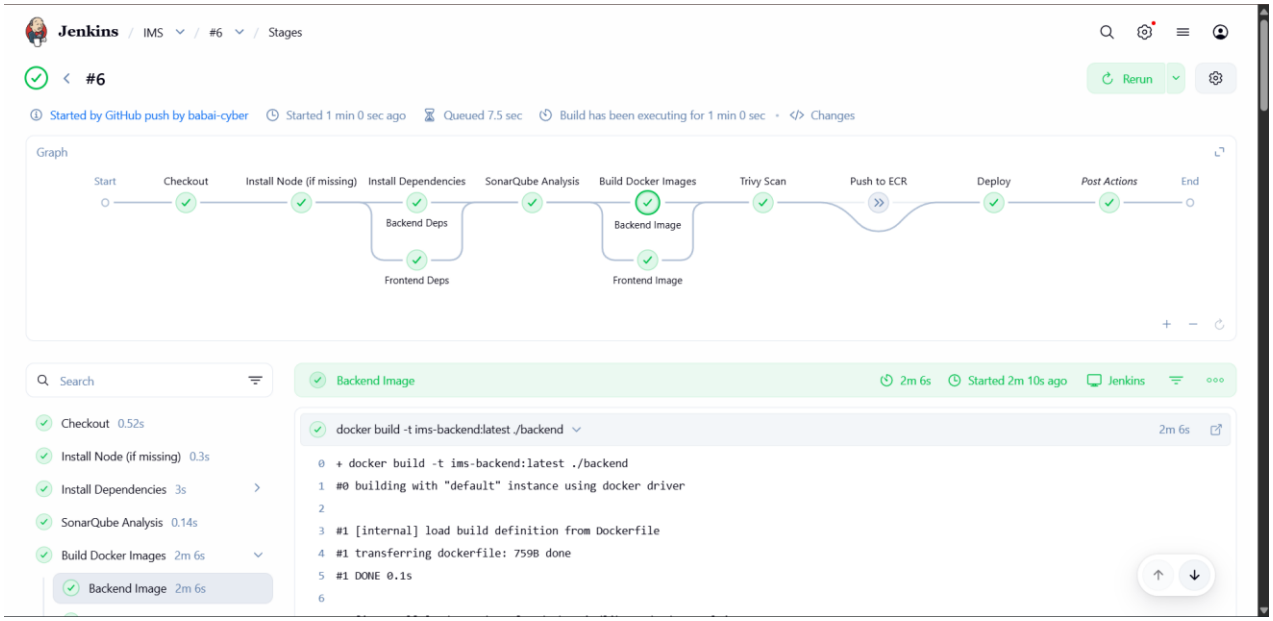
**Frontend (React + Vite) → Backend (Node.js + TypeScript + Express) → PostgreSQL (Work Items + RCA) + MongoDB (Raw Signals) + Redis (Cache + Queue)** All services are containerized with **Docker Compose** and orchestrated via **Jenkins CI/CD**. Monitoring is powered by **Prometheus + Grafana**.

## 4. Architecture Explanation

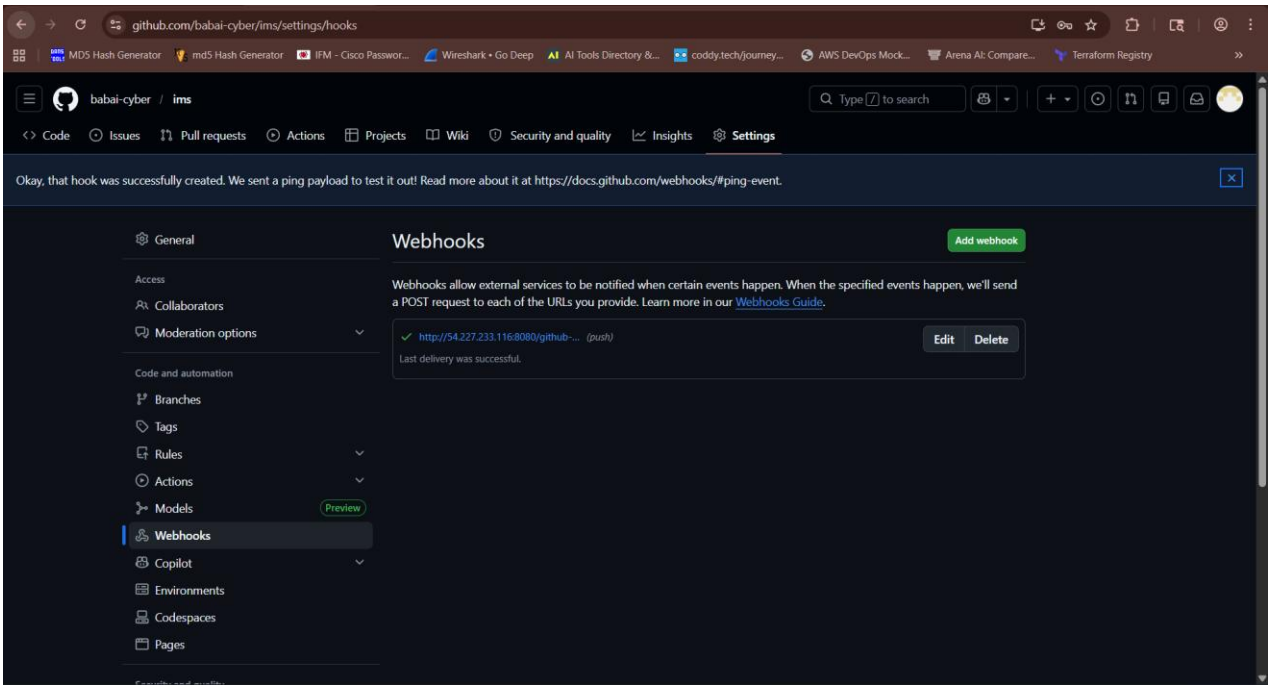
- **Frontend**: React dashboard auto-refreshing every 5s, showing incidents and RCA forms.
- **Backend**: Node.js API with Bull queue for backpressure, Redis debounce logic, PostgreSQL for transactional state, MongoDB for raw signals.
- **Databases**: PostgreSQL (ACID transactions), MongoDB (TTL audit log), Redis (cache + queue).
- **Design Patterns**:
    - Strategy Pattern → P0 alerts for RDBMS/API, P2 alerts for Cache/Queue. ○ State Machine → Enforces OPEN → INVESTIGATING → RESOLVED → CLOSED lifecycle.

## 5. CI/CD Pipeline (Jenkins)

**Before webhook trigger :**




After Webhook trigger:

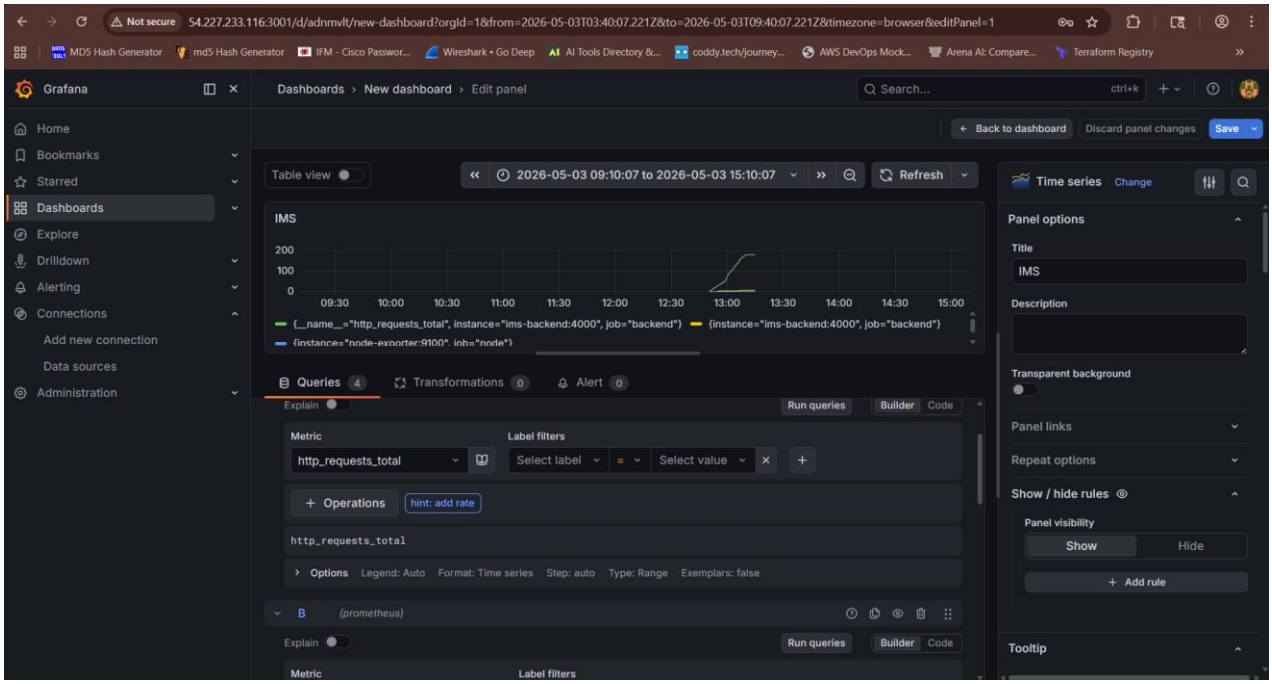



Stages:

1. **Checkout** (GitHub webhook) : https://github.com/babai-cyber/ims.git
2. **Install Dependencies** (npm ci)
3. **Build** (TypeScript compile)
4. **Unit Tests** (Jest for RCA/state machine/debounce)
5. **SonarQube Analysis** (Quality Gate enforcement)
6. **Trivy Scan** (Docker image CVE scanning)

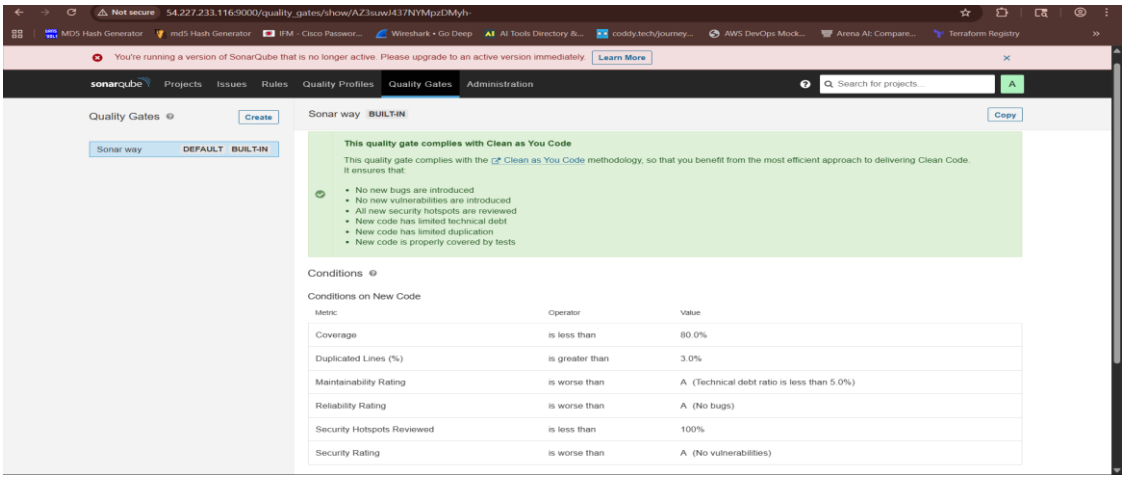
7. **Docker Build & Push** (Docker Hub/ECR)
8. **Deploy** (EC2 via docker-compose)
9. **Health Check** (/health endpoint)

## 6. Monitoring (Prometheus + Grafana)



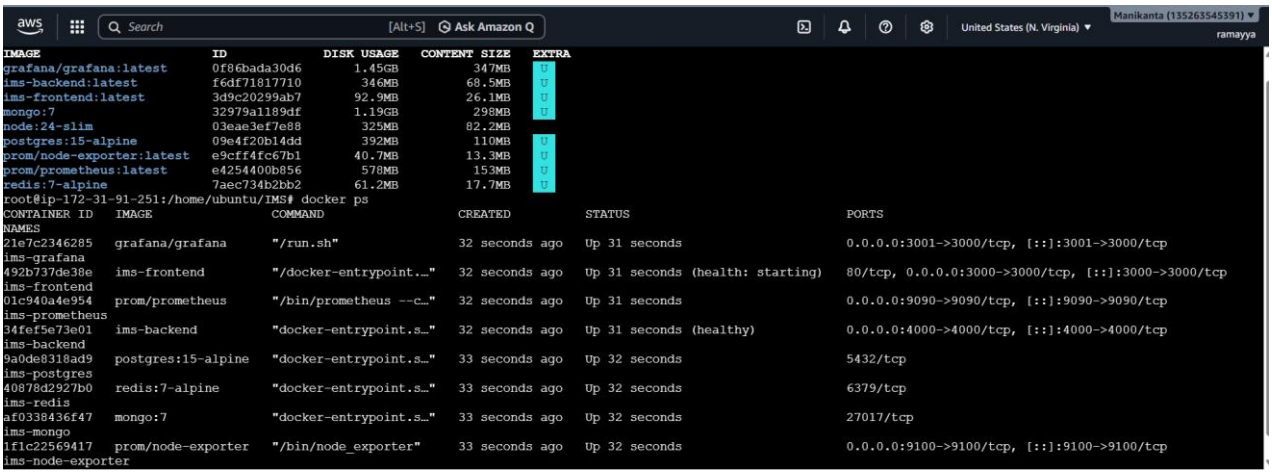

- **Prometheus** scrapes metrics every 15s from backend and node-exporter.
- **Grafana Dashboard Panels**:
  - Signal Throughput (rate of signals/sec) ○ Queue Depth (Bull jobs waiting) ○ HTTP Latency (P99 histogram) ○ Open Incidents Count

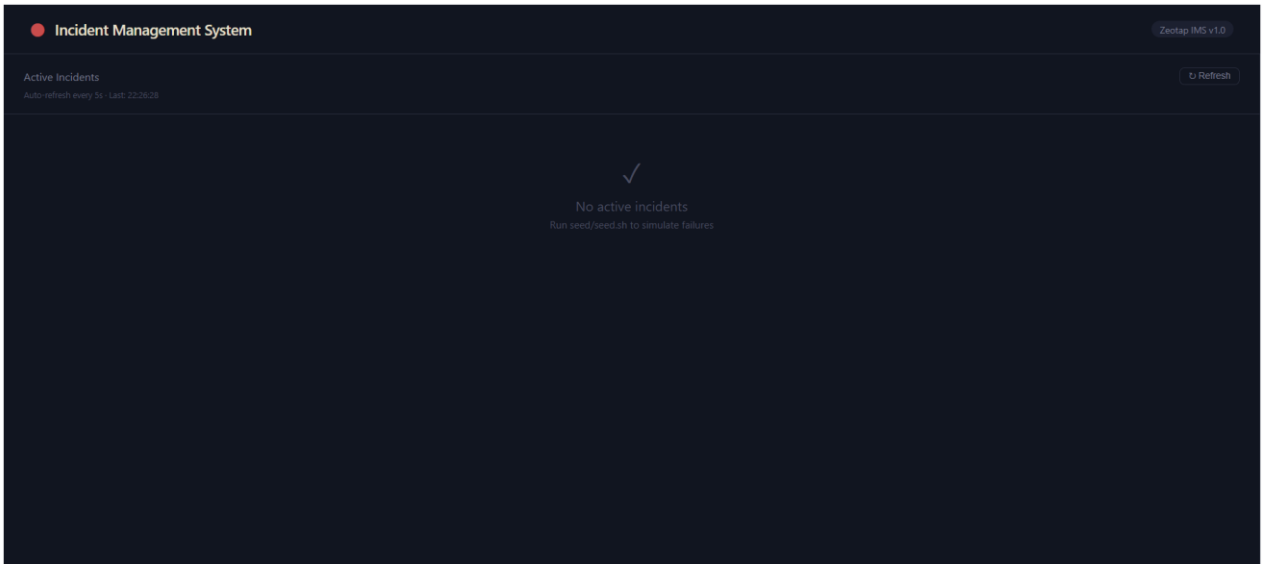
## 7. Security & Best Practices



- **Rate Limiting**: 10k requests/min per IP.
- **Input Validation**: Required fields enforced at API level.
- **CORS + Helmet.js**: Secure headers, restricted origins.
- **Secrets Management**: Environment variables + Jenkins credentials store.
- **Trivy CVE Scanning**: Fail pipeline on CRITICAL vulnerabilities.
- **SonarQube SAST**: No blockers/critical hotspots.
- **MongoDB TTL Index**: Auto-delete signals after 90 days.
- **AWS Security Groups**: Only required ports exposed.

## 8. Deployment




- **Local**: docker-compose up --build
- **Cloud**: AWS EC2 (t3.medium) .
- **Health Check**: curl http://<ec2-ip>:4000/health
- **Frontend**: http://<ec2-ip>:3000

- **Monitoring**: Prometheus (:9090), Grafana (:3001).

**9. Screenshots (Attached in Submission)** link for project screenshots

- Application UI (Incident Dashboard)
- Jenkins Pipeline Execution
- Grafana Dashboard
- Prometheus Targets
- SonarQube Quality Gate Report

**10. Conclusion**

This project demonstrates **SRE capabilities**:

- Handling **10k signals/sec** with backpressure.
- Enforcing **debounce logic** and RCA gating.
- Full **DevSecOps pipeline** with SonarQube + Trivy.
- **Observability stack** with Prometheus + Grafana.
- Secure, resilient, and scalable deployment on Docker + AWS EC2.